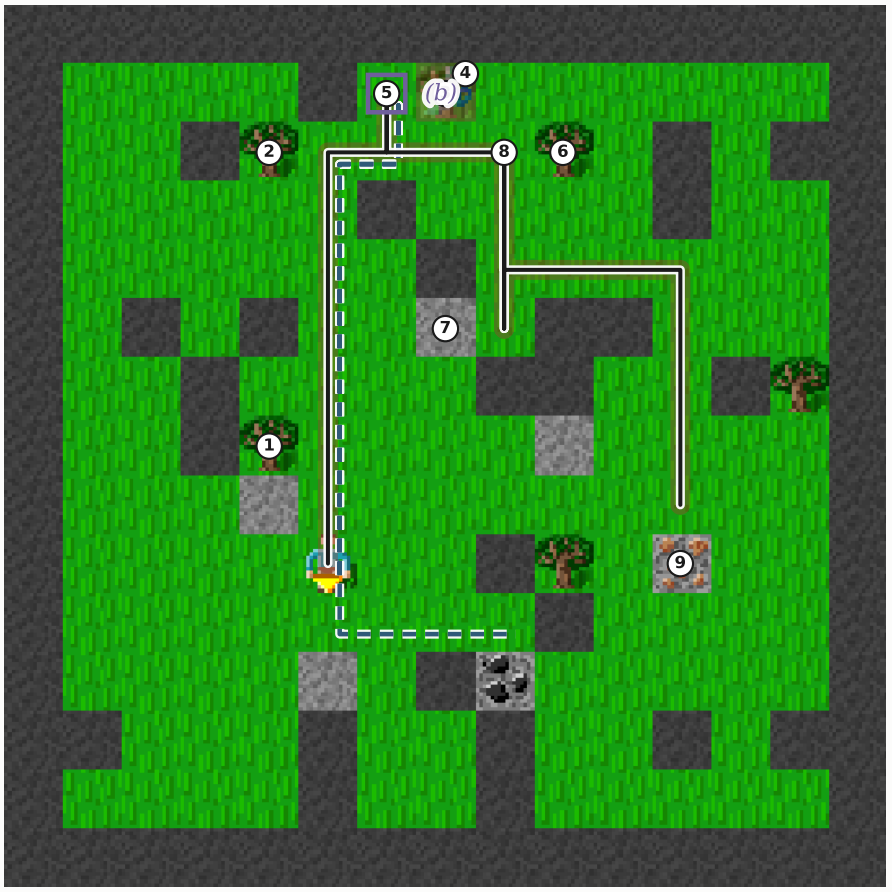
(a) One held-out field, and what the policy did on it

iron 1.1, coal 0.5, 0.05 per action · iron chain 40, coal chain 32: a detour of 8, under 12, so iron

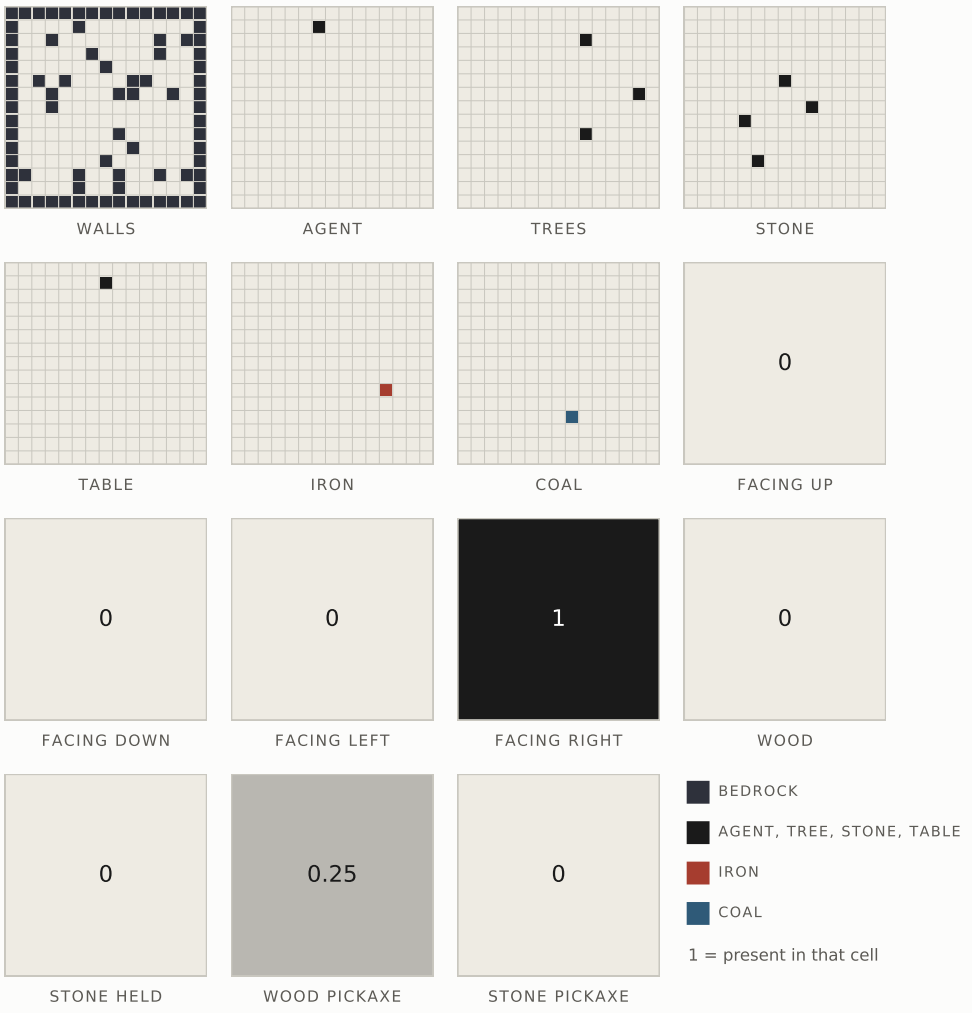


EXPERT'S IRON PLAN · 40 — EXPERT'S COAL PLAN · 32 — POLICY'S ROLLOUT · 40 ACTIONS

① tree cut · step 4
② tree cut · step 11
③ tree cut · step 15
④ table placed · step 16
⑤ wood pickaxe · step 17
⑥ tree cut · step 21
⑦ stone mined · step 26
⑧ stone pickaxe · step 30
⑨ iron mined · step 40

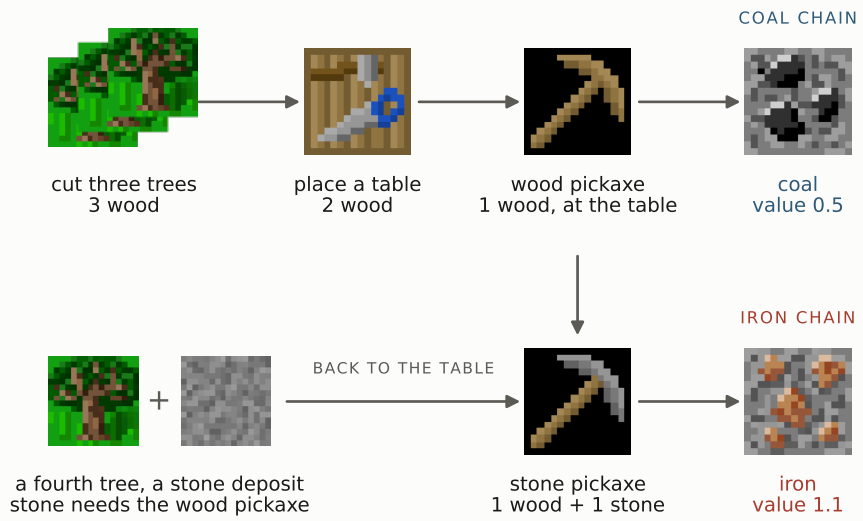
(b) What the policy sees at step 17

table down, wood pickaxe in hand · map planes, then facing and inventory as constant planes



(c) The tech tree the expert plans over

the iron chain is the coal chain plus a detour for its ingredients



The expert plans both chains from the field, prices each at value minus 0.05 per action, and takes the better one. Iron is worth 0.6 more than coal, so its chain wins whenever its detour costs fewer than 12 actions. Every field is a different price for the same two goods.

(d) The decision rule over the held-out fields

take iron iff its detour costs fewer than (1.1 − 0.5) / 0.05 = 12 actions

